

ECE 332 Spring 2026

Exam 2 Solutions

(100 pts total)

June 9, 2026

Allowed: Calculator approved for the FE exam and a 2-sided page of notes.

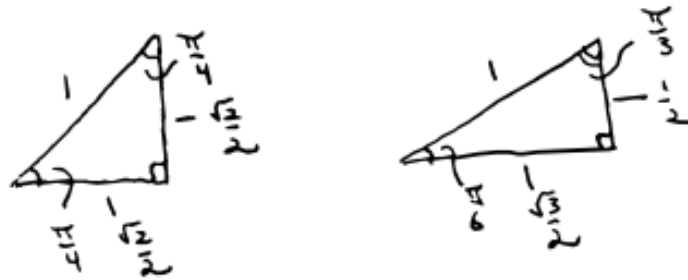
Not allowed: Any other devices or materials.

Use the approximations

$$\epsilon_0 \approx \frac{1}{36\pi} \times 10^{-9} \text{ F/m}, \quad \mu_0 = 4\pi \times 10^{-7} \text{ H/m}, \quad \eta_0 \approx 120\pi \, \Omega$$

in your calculations and **include the units** in all answers.

Helpful Fact: The wavelength of an EM wave is $\lambda = u_p/f$.



1. Two lossless, nonmagnetic media meet at the x - y plane with Medium 1 filling the half space $z \leq 0$ and Medium 2 filling the half space $z \geq 0$. Here, $\epsilon_{r1} = 9$ and $\epsilon_{r2} = 16$. A 200 MHz x -polarized EM wave traveling in Medium 1 with electric field phasor

$$\tilde{\mathbf{E}} = 6e^{-jk_1 z} \hat{\mathbf{x}} \text{ (V/m)}$$

meets the boundary between the media at normal incidence.

(24 points total)

- (a) Find the angular frequency ω of the EM wave and the wavenumber k_2 of the EM wave in Medium 2. (5 points)

The angular frequency of the EM wave is

$$\omega = 2\pi f = (2\pi \text{ rad})(2 \times 10^8 \text{ Hz}) = 4\pi \times 10^8 \text{ rad s}^{-1}$$

and the wavenumber in Medium 2 is

$$k_2 = \frac{\omega}{c} \sqrt{\epsilon_{r2}} = 4 \left(\frac{4\pi \times 10^8 \text{ rad s}^{-1}}{3 \times 10^8 \text{ m s}^{-1}} \right) = \frac{16\pi}{3} \text{ rad m}^{-1}.$$

- (b) Find the reflection coefficient Γ and the transmission coefficient τ . (6 points)

The reflection coefficient is

$$\Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{3 - 4}{3 + 4} = -\frac{1}{7}$$

and the transmission coefficient is

$$\tau = 1 + \Gamma = \frac{6}{7}.$$

- (c) Find the phasor of the electric field of the transmitted wave. (5 points)

The phasor of the electric field of the transmitted wave is

$$\tilde{\mathbf{E}}^t = \tau E_0^i e^{-jk_2 z} \hat{\mathbf{x}} = \left(\frac{6}{7} \right) (6 \text{ V/m}) e^{-j(16\pi/3)z} \hat{\mathbf{x}} = \left(\frac{36}{7} \text{ V/m} \right) e^{-j(16\pi/3)z} \hat{\mathbf{x}}.$$

- (d) Determine the average power density in Medium 1. (8 points)

The intrinsic impedance of Medium 1 is

$$\eta_1 = \frac{\eta_0}{\sqrt{\epsilon_{r1}}} = \frac{120\pi \Omega}{3} = 40\pi \Omega.$$

and the average power density in Medium 1 is

$$\begin{aligned} \mathbf{S}_{\text{av1}} &= \frac{|E_0^i|^2}{2\eta_1} (1 - |\Gamma|^2) \hat{\mathbf{z}} = \frac{36}{80\pi} \left(1 - \frac{1}{49} \right) \hat{\mathbf{z}} (\text{W/m}^2) \\ &= \frac{9}{20\pi} \left(\frac{48}{49} \right) \hat{\mathbf{z}} (\text{W/m}^2) = \frac{108}{245\pi} \hat{\mathbf{z}} (\text{W/m}^2) \\ &\approx 0.14032 \hat{\mathbf{z}} (\text{W/m}^2). \end{aligned}$$

2. Suppose two lossless, nonmagnetic media meet at the x - y plane with Medium 1 filling the half space $z \leq 0$ and Medium 2 filling the half space $z \geq 0$. Here, $\epsilon_{r1} = 9$ and $\epsilon_{r2} = 16$. (18 points total)

- (a) Find the indices of refraction for the two media. (6 points)

The index of refraction for Medium 1 is

$$n_1 = \sqrt{\mu_{r1}\epsilon_{r1}} = \sqrt{\epsilon_{r1}} = 3$$

and the index of refraction of Medium 2 is

$$n_2 = \sqrt{\mu_{r2}\epsilon_{r2}} = \sqrt{\epsilon_{r2}} = 4.$$

- (b) Is there a critical angle in this case? Explain why or why not. If there is a critical angle, find its sine. (4 points)

Since $n_2 > n_1$, there is no critical angle in this case.

If an incident EM wave meets the boundary at $\theta_i = \pi/6$ rad,

- (c) Find the angle of reflection. (4 points)

From Snell's law of reflection,

$$\theta_r = \theta_i = \pi/6 \text{ rad.}$$

- (d) Find the sine of the angle of transmission. (4 points)

From Snell's law of refraction,

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i = \frac{3}{4} \left(\frac{1}{2} \right) = \frac{3}{8} = 0.375.$$

3. Suppose two lossless, nonmagnetic media meet at the x - y plane with Medium 1 filling the half space $z \leq 0$ and Medium 2 filling the half space $z \geq 0$. Here, $\epsilon_{r1} = 9$ and $\epsilon_{r2} = 16$. Furthermore, suppose an EM plane wave meets the boundary at an angle $\theta_i \neq 0$. (12 points total)

- (a) Find the tangent of the Brewster angle for the component of the wave that is polarized parallel to the plane of incidence. (4 points)

Since the media are nonmagnetic,

$$\tan \theta_{B\parallel} = \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} = \frac{4}{3} = 1.3333.$$

If the angle of incidence is the Brewster angle for the component of the wave that is polarized parallel to the plane of incidence,

- (b) What is the polarization of the reflected wave? Explain. (4 points)

At this angle of incidence, the reflected wave is polarized perpendicular to the plane of incidence.

- (c) What is the polarization of the transmitted wave? Explain. (4 points)

The transmitted wave is the superposition of a wave polarized parallel to the plane of incidence and a wave polarized perpendicular to the plane of incidence.

4. Consider an air-filled, metal, rectangular waveguide with width $a = 6$ cm and height $b = 3$ cm. (14 points total)

- (a) What is the dominant TE mode for the waveguide? (4 points)

The dominant TE mode is TE_{10} .

- (b) What is the lowest frequency for an EM wave that will propagate through the waveguide? (5 points)

The lowest frequency for an EM wave that will propagate through the waveguide is the cutoff frequency for the dominant TE mode, i.e. for TE_{01} . This frequency is

$$f_{10} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{u_{p0}}{2a}.$$

Since the waveguide is air-filled, $u_{p0} = c$ and

$$f_{10} = \frac{c}{2a} = \frac{3 \times 10^8 \text{ m s}^{-1}}{0.12 \text{ m}} = 2.5 \times 10^9 \text{ Hz} = 2.5 \text{ GHz}.$$

- (c) If the open ends of the waveguide are closed with additional metal walls that make the length of the resulting cavity 8 cm, what frequencies will persist in the cavity for the mode TM_{110} ? (5 points)

The only frequency that will persist for the TE₁₁₀ mode is the resonant frequency

$$\begin{aligned} f_{110} &= \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2} = \frac{c}{2} \sqrt{\left(\frac{1}{0.06 \text{ m}}\right)^2 + \left(\frac{1}{0.03 \text{ m}}\right)^2} \\ &= \frac{3 \times 10^8 \text{ m s}^{-1}}{0.06 \text{ m}} \sqrt{\frac{1}{4} + 1} = 2.5\sqrt{5} \times 10^9 \text{ s}^{-1} \\ &= 2.5\sqrt{5} \text{ GHz} \approx 5.5902 \text{ GHz}. \end{aligned}$$

5. Suppose that a dipole antenna with directivity $D = 3\pi/2$ is transmitting a 8 GHz EM wave into air. (20 points total)

- (a) Find the wavelength of the emitted EM wave. (5 points)

The wavelength is

$$\lambda = \frac{u_p}{f} = \frac{c}{f} = \frac{3 \times 10^8 \text{ m s}^{-1}}{8 \times 10^9 \text{ s}^{-1}} = 0.0375 \text{ m} = 3.75 \text{ cm}.$$

- (b) Find the effective area of the antenna for the emitted radiation. (5 points)

The effective area of the antenna at this wavelength is

$$\begin{aligned} A_e &= \frac{\lambda^2 D}{4\pi} = \frac{(0.0375)^2 (3\pi/2)}{4\pi} \text{ m}^2 = \frac{3}{8} (0.0375)^2 \text{ m}^2 \\ &= 5.2734375 \times 10^{-4} \text{ m}^2. \end{aligned}$$

- (c) What is the physical area of the aperture a large-aperture antenna with the same directivity at this wavelength? (5 points)

For a large-aperture antenna with the same directivity, the area of the physical aperture for this wavelength would be

$$A_p \approx A_e = 5.2734375 \times 10^{-4} \text{ m}^2.$$

- (d) If the aperture is circular, would the aperture field antenna truly be a large-aperture antenna at this wavelength? Explain. (5 points)

The linear dimension that characterizes the aperture is its diameter d . If r is the radius of the aperture, the area of the aperture is

$$\begin{aligned} A_p &= \pi r^2 = \pi(d/2)^2 = \pi d^2/4 \\ &\implies \\ d &= 2\sqrt{A_p/\pi} \approx 0.025912 \text{ m.} \end{aligned}$$

Since this is smaller than the wavelength, this would not be a large-aperture antenna at this wavelength.

6. Consider a linear phased array oriented along the z -axis consisting of 4 half-wave dipole antennas intended for transmitting 96 MHz EM waves into air, where neighboring antennas are separated by 16λ . (12 points total)

- (a) What incremental phase delay would be needed to steer the maximum value of the array factor to the end-fire direction? (4 points)

The scan angle to steer the array to the end-fire direction is $\theta_0 = 0$. So, in this case, the incremental phase delay is

$$\delta = kd \cos \theta_0 = kd \cos 0 = kd = \frac{2\pi d}{\lambda} = \frac{2\pi(16\lambda)}{\lambda} = 32\pi.$$

- (b) Find the incremental phase delay needed to steer the maximum value of the array factor to the scan angle $\pi/3$ rad. (4 points)

The incremental phase delay that steers the array to the scan angle $\pi/3$ rad is

$$\delta = kd \cos \theta_0 = kd \cos(\pi/3) = \frac{2\pi d}{\lambda} \cos(\pi/3) = 32\pi(1/2) = 16\pi.$$

- (c) If the average power density of a half-wave dipole antenna in the x - y plane is

$$S_0 = \frac{15I_0^2}{\pi R^2}$$

and all of the amplitudes are $a = 1/5$, what is the power density in the x - y plane in the direction of the scan angle? (4 points)

The average power density of the array is

$$S(R, \theta, \phi) = S_e(R, \theta, \phi)F_a(\theta),$$

where S_e is the power density of a half-wave dipole antenna and F_a is the array factor. Since the amplitudes are equal, the array factor has its maximum value $F_{a,\max} = a^2 N^2 = 1$ in the direction of the scan angle in the x - y plane. So, in the direction of the scan angle in the x - y plane, the power density of the array is

$$S = S_0 F_{a,\max} = S_0 = \frac{15I_0^2}{\pi R^2}.$$